2026 Kentucky Cybersecurity and Forensics Conference

BOOK OF ABSTRACTS

Friday, October 30, 2026

Harold R. Benson Research and Demonstration Farm

Building Cyber Resilience Through Collaboration: Academia, Industry, and Government

Hosted by

Kentucky State University

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Session: WELCOME AND LOGISTICS | 08:00 – 08:30 am

Dr. Koffi C. Akakpo
President of Kentucky State University
Kentucky State University

BRIEF BIOGRAPHY

Dr. Koffi Akakpo is an enthusiastic leader who advocates for student success on all levels. His "student-first" philosophy is the cornerstone of his work in higher education.

He became the 19th president of Kentucky State University on July 1, 2023, bringing to the role more than two decades of leadership in academic innovation, fiscal stewardship, and student success.

At Kentucky State, he is leading a new era of progress shaped by *Vision 2028: Thorobreds Leading for Impact*, a comprehensive strategic plan designed to expand access, strengthen student achievement, and deepen the University's service to the Commonwealth and beyond. His leadership also advances the University's 1890 land-grant mission, ensuring that research and Cooperative Extension programs deliver meaningful solutions in agriculture, health, and natural resources.

Prior to joining Kentucky State, Dr. Akakpo served as president of Bluegrass Community and Technical College, where he oversaw record enrollment growth and launched initiatives to improve advising, retention, and workforce alignment. He previously held senior leadership positions at Central State University, an 1890 land-grant HBCU in Ohio, and at North Central State College, where his financial leadership helped restore institutional stability. He also served with the Ohio Department of Natural Resources.

Dr. Akakpo earned his Ph.D. in higher education administration from the University of Toledo and his master's degree in business administration from Ashland University.

Session: KEYNOTE ADDRESS | 08:45 – 09:30 am

Title: Operation Angel – Prepare, Prevent, Provide

Mr. Rick Nord
Regional Security Specialist
US Secret Service / Kentucky Office of the Attorney General, Counter Exploitation

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ABSTRACT

The digital environment has transformed the ways children are targeted, exploited, and coerced. This keynote examines the growing intersection of online child sexual exploitation, human trafficking, generative artificial intelligence (GAI), encrypted communications, and platform-enabled anonymity. It will explore how tactics such as sextortion, the nonconsensual exchange of intimate images, content obfuscation, and end-to-end encryption can complicate prevention, identification, reporting, and investigative efforts.

The presentation will highlight the importance of prevention, education, and coordinated community response. A central focus will be the Operation Angel Summit, an interactive event presented through a partnership among National Safe Place, the United States Secret Service, and the Commonwealth of Kentucky Office of the Attorney General. Designed to educate and empower youth, the Summit provides critical information on prevention, victim resources, and survivor support. Through engaging presentations and discussions, young people gain tools to recognize risks, protect themselves and their peers, and take action when they encounter exploitation or crisis situations.

Participants will leave with a clearer understanding of the current threat landscape, emerging challenges, and the shared responsibility to protect children through prevention, early intervention, victim-centered support, and effective collaboration.

BRIEF BIOGRAPHY

Mr. Rick Nord is a Regional Security Specialist and former Supervisory Special Agent with the United States Secret Service, bringing approximately 25 years of experience in law enforcement, cybersecurity, and child protection. He previously supervised the Kentucky Cyber Fraud Task Force and founded Operation Angel, an initiative focused on preventing child exploitation through education and collaboration. He currently works in counter-exploitation outreach with the Kentucky Office of the Attorney General. He also serves as an Ambassador for the National Center for Missing & Exploited Children and is involved with organizations dedicated to youth safety and protection. His work focuses on cybersecurity awareness, crime prevention, and protecting vulnerable communities.

Session: CYBERSECURITY IN AI | 09:45 – 10:30 am

Title: The Cybersecurity Paradox: AI as Defender and Threat

Dr. Adel Elmaghraby
Winnia Professor of Computer Science and Engineering and Director of Industrial Research and Innovation
J.B. Speed School of Engineering, University of Louisville

ABSTRACT

Artificial intelligence is transforming how we work, communicate and make decisions, and cybersecurity is no exception. The same technology that helps detect threats, identify vulnerabilities and accelerate incident response also enables convincing phishing, automated attacks and sophisticated deception, including synthetic voice and video that can impersonate a trusted colleague. This is the cybersecurity paradox: AI strengthens digital defenses while creating new opportunities for attackers.

The paradox is no longer theoretical. Industry reports from 2026 describe a sharp rise in AI-enabled attacks, and researchers have documented espionage campaigns in which AI performed most of the technical work. Meanwhile, defenders use AI to correlate alerts, summarize investigations, prioritize risks and find and patch flaws at new scale. More advanced systems are also finding and exploiting vulnerabilities in software at every level, shrinking the time defenders must respond.

This presentation examines the benefits and limits of AI-driven security, from machine learning to generative AI to agents that act through connected tools. It also explores the risks of adversarial manipulation, including data poisoning, evasion and prompt injection, and of overreliance on automated systems, illustrated with recent real-world cases.

BRIEF BIOGRAPHY

Dr. Adel Elmaghraby is the Winnia Professor of Computer Science and Engineering and Director of Industrial Research and Innovation at the University of Louisville's J.B. Speed School of Engineering. His research spans artificial intelligence, cybersecurity, intelligent multimedia systems, and high-performance computing, with applications in healthcare, smart cities, and data analytics. He earned his Ph.D. in Electrical Engineering from the University of Wisconsin–Madison and has held appointments there and at Carnegie Mellon University's Software Engineering Institute.

Dr. Elmaghraby combines research, industry collaboration, and workforce development to advance AI and cybersecurity. He is a co-principal investigator on UofL's NSA-funded healthcare cybersecurity initiative, and the NSF Cybersecurity Scholarship for Service (SFS). He is the former department chair, and the current advisor for the cybersecurity graduate certificate, and he has advised more than 60 master's and 43 doctoral graduates and is an IEEE Life Senior Member. His work brings a broad perspective on how AI can strengthen cybersecurity while creating new challenges for protecting digital systems.

Session: INDUSTRY PANEL | 10:40 – 11:50 am

Dr. Shuangteng Zhang
Director of the Digital Forensics and Cybersecurity Institute

Eastern Kentucky University

BRIEF BIOGRAPHY

Dr. Shuangteng Zhang is a Professor in the Department of Computer Science and Information Technology at Eastern Kentucky University (EKU) and serves as Director of the Digital Forensics and Cybersecurity Institute. With over 20 years of service at EKU, he has played a key role in advancing computing education through curriculum development, program innovation, and student mentorship.

He was instrumental in the creation of EKU's Bachelor of Science degree program in Digital Forensics and Cybersecurity, helping to establish a robust foundation for cybersecurity education at the university. As Director, Dr. Zhang oversees program coordination, teaches core courses, and mentors students in research and career development. He also fosters collaboration with faculty and industry partners to strengthen the institute's academic and applied mission.

Dr. Zhang earned his Ph.D. in Electrical Engineering from the University of Toledo. His research interests include digital forensics, cybersecurity, signal and image processing, and machine learning. He has published and presented in these areas and continues to support applied research and student engagement across cybersecurity disciplines.

Dr. Shishir Shukla
Data, Analytics & AI Leader

Thermo Fisher Scientific

BRIEF BIOGRAPHY

Dr. Shishir Shukla is an experienced Data, Analytics, and Artificial Intelligence leader at Thermo Fisher Scientific, specializing in data management, business intelligence, analytics, and automation. He holds a Ph.D. and has a professional background in developing enterprise data solutions and implementing advanced analytics technologies. His expertise includes data governance, business intelligence platforms, and digital transformation. He previously worked at Schneider Electric, where he contributed to implementing enterprise analytics and data management platforms. Dr. Shukla is passionate about leveraging data and emerging technologies to improve organizational efficiency, support informed decision-making, and drive innovation.

Mr. Jeff Jareki
Chief Information Security Officer
Volta, Inc

BRIEF BIOGRAPHY

Jeff has worked in technology for over 25 years. He began his career in IT helpdesk where he learned the ins/outs of user interfaced systems. He then moved into a more traditional IT analyst role. Jeff took his passion for programming into a position as a software engineer where he developed many client-facing systems that were deployed in the ancient days of CD distribution.

16 years ago, he accepted his first position in cyber security as an analyst. Utilizing his IT/programming background, he was able to automate many security controls and spent additional time beginning his aggressive cyber security certification paths. The first certification he earned was the CISSP. In the past 16 years, he's earned many more cyber security certifications, and began teaching boot camps helping other security professionals earn theirs.

Jeff has worked in multiple domains of security, including, Identity and Access Management, Audit (internal and external), Architecture, Enterprise and Third Party Risk for both core business as well as Mergers and Acquisitions and Cyber Security Leadership.

Mr. Colton Tate
Sr. Account Executive
CanIPhish

BRIEF BIOGRAPHY

Mr. Colton Tate is a Senior Account Executive at CanIPhish, a cybersecurity company specializing in phishing simulation and security awareness training. His professional role focuses on connecting organizations with cybersecurity solutions that help strengthen employee awareness and reduce risks associated with phishing and social engineering attacks. He is participating as an industry panel speaker at the 2026 Kentucky Cybersecurity and Forensics Conference, contributing to discussions on cybersecurity challenges and industry solutions.

Mr. James Meece
Chief Information Security Officer
Louisville Metro Government

BRIEF BIOGRAPHY

James has served as the Chief Information Security Officer for Louisville Metro Government since 2015. During his tenure, he and his team have led the charge in reducing overall cyber risk to Louisville Metro by 90% through metrics-driven action. He has embraced an approach of progress through partnerships - both internal and external - and forged a strategy based on defense in depth that adapts to the disappearing network perimeter. He is a retired Major for the Kentucky National Guard and served in the military for 33 years, including a mobilization with US Cyber Command in 2022 as a Cyber and Electronic Warfare Officer. He has been inducted into the Honorable Order of Saint Barbara (Field Artillery Association), the Honorable Order of Thor (Cyber Corps Association), and the Honorable Order of Kentucky Colonels. He also holds multiple industry certifications and is a member of the Association of Old Crows, a member the Kentucky Cyber Threat Working Group, and a founding member and President of the Kentucky Cyber Intelligence Information Initiative (KCI3)

Mr. Bob Stamper
Chief Executive Officer
iSAFE Complete Managed Service

ABSTRACT

iSAFE Complete helps people use technology to reach their goals while protecting their information and hardware through education, proactive support, and software solutions. For more than 19 years, the company has served the technology needs of Central Kentucky with the belief that technology should work for you. Its Complete Managed Services, VoIP, and Complete Web Presence packages give growing businesses what they need to improve productivity, strengthen security, and manage costs. iSAFE believes that the more productive a team is, the more profitable the business becomes, and it puts long-term working relationships ahead of short-term sales goals. In 2025, iSAFE Complete was named a finalist in the MSP Titans of the Industry awards.

BRIEF BIOGRAPHY

Bob Stamper is the Owner and CEO of iSAFE Complete, where he has spent more than 25 years helping protect businesses across Kentucky. He holds the Certified Information Systems Security Professional (CISSP) certification, serves as an ambassador for the Richmond Chamber of Commerce, Commerce Lexington, was named the 2023 Ambassador of the Year in Richmond and was recognized as one of the Top 25 Leaders in Madison County in 2025. Internationally recognized for excellence in leadership, service, and growth, iSAFE Complete was named a finalist in the MSP Titans of the Industry awards in 2025.

Ms. Sheri Donahue

CEO & Co-Founder

Commonwealth Sentinel Cyber Security

BRIEF BIOGRAPHY

Sheri Donahue is an engineer and former U.S. Navy Program Manager for Security and Intelligence, where she led intelligence-driven technology programs supporting the Department of Defense and national security agencies. She is President-Emeritus of the InfraGard National Members Alliance, having served 12 years on the National Board and three as National President, where she strengthened public-private partnerships with the FBI and DHS and created the InfraGard Sector Chief Program now used nationwide.

Sheri has led cyber policy and research initiatives as President & Executive Director of the Cyber Conflict Studies Association and later directed strategic cybersecurity partnerships for Humana. She has also worked across healthcare incident response, national critical infrastructure protection, and state-level cyber policy, including running for Kentucky State Auditor on a cybersecurity-focused platform.

She is a frequent speaker, published author, and recipient of multiple awards from the Navy, FBI, and Army, including InfraGard's first national Lifetime Leadership Award. Sheri holds a BS in Industrial Engineering from Purdue University and currently serves part-time as Executive Director of Friend for Life Cancer Support Network in addition to her role as CEO of Commonwealth Sentinel.

Elwyn Rainer II

CEO | Security Operations Manager

Elwyn Rainer 2 LLC | KY State Government

BRIEF BIOGRAPHY

Elwyn Rainer II is an experienced cybersecurity professional, three-time published author, keynote speaker, and leadership development professional. With extensive experience across government and private-sector environments, he specializes in cybersecurity strategy, security operations, governance, risk management, compliance, and incident response.

As CEO of Elwyn Rainer 2 LLC and Desire Better Solutions LLC, Elwyn is passionate about developing future leaders, strengthening cybersecurity awareness, and helping individuals and organizations reach their full potential.

His work bridges cybersecurity, workforce development, career readiness, and effective leadership. Through speaking engagements, mentorship, and professional development initiatives, he encourages individuals to take ownership of their growth, embrace continuous learning, and develop the skills necessary to succeed in an evolving workforce.

Session: HANDS-ON SESSION A | 01:15 – 02:15 pm

Dr. Adrian Lauf
Associate Professor of CS and Engineering
University of Louisville

BRIEF BIOGRAPHY

Dr. Adrian Lauf is an associate professor in the Computer Science & Engineering department at the University of Louisville's J. B. Speed School of Engineering. He is a core faculty member at the Louisville Automation and Robotics Research Institute (LARRI) as a part of the Autonomous Intelligent Mobile Systems (AIMS) lab. Dr. Lauf's research expertise is in cooperative autonomous aerial robotics, which has the goal of enabling swarms of autonomous UAVs to accomplish objectives in a cooperative and collaborative manner. These robot swarms use multi-domain mesh networking, decentralized decision-making, and node clustering to improve communication efficiency. Dr. Lauf also has research interests in cyber and network security, with research efforts in embedded and IoT security, as well as several projects in cybersecurity workforce development.

Dr. Lauf also is the director of the CyberEDU cybersecurity education cluster, which provides experiential learning for students by simulating real networking environments and security tools to train the next generation of cybersecurity professionals and researchers. Dr. Lauf is the point of contact for the University of Louisville's National Center of Academic Excellence in Cyber Defense (NCAE-CD). He is also a member of the HIVE AI Innovation studio and an affiliate member of the Logistics and Distribution Institute at the University of Louisville.

Session: HANDS-ON SESSION B | 01:15 – 02:15 pm

Title: Case Closed: A Hands-On Digital Forensics Investigation

Dr. Cheryl Purdy
Digital Forensic Examiner/Analyst
University of the Cumberlands

ABSTRACT

A burglary has been reported, a family heirloom is missing, and a suspect's phone has just been seized. Can you find the evidence hiding inside it?

In this hands-on session, participants step into the role of a digital forensic examiner and work a realistic mock case from start to finish. Using Autopsy, a free, widely used forensic tool, attendees will examine a forensic image of an Android phone to uncover photos of stolen property, messages arranging a sale, contact records, and the hidden details stored inside digital files. Along the way, they'll learn how investigators find files that aren't what they appear to be, and use location and time data in photos to connect separate pieces of evidence. No prior forensics experience is needed.

BRIEF BIOGRAPHY

Dr. Cheryl Purdy is a digital forensics examiner/analyst with the Davies County Sheriff's Office in Owensboro Kentucky. She holds several industry certifications in the field and has been with the department in this capacity for the past 13 years. Cheryl holds a PhD in Information Technology with a specialty in Digital Forensics, an MBA with concentration in Computer Information Systems, and a Bachelor of Science Degree in Mathematics. She has completed training from the National White Collar Crime Center, Secret Service's National Computer Forensics Institute, Boston University Center for Cybercrime, Investigation, and Cybersecurity, and the American Academy of Applied Forensics.

In addition to her position as a digital forensic analyst, Dr. Purdy teaches graduate level digital forensic courses for the University of the Cumberlands, Network Administration and Information Security for Western Kentucky University, and a workforce Cybersecurity course for the University of Louisville. She has recently returned from Iceland as a Fulbright Specialist working with the University of Akureyri in the development of cybersecurity curriculum for law enforcement.

Dr. Purdy received the 2023 Sheriff's Excellence Award outlining sentences of over 200 years in 5 major cases. She also received the 2021 National Sheriff's Department Reserve Deputy of the year award, the 2021 New Beginnings (Sexual Assault Support Services) Voice for Change Award, and the 2021 Owensboro Noon Optimist Respect for Law Award. In addition, she is a member of the Kentucky Office of Homeland Security Cyber Threats Working Group.

A few of the presentations she has given include *The Evolution of Digital Forensic in Criminal Investigations* and *Digital Anatomy of a Homicide* at the University of Akureyri; *Digital Forensics and Information Security* instructional materials to Kosovo Academy of Public Safety; and *Ethics in the Field of Digital Forensics* at Kentucky Prosecutors Conference.

Session: CYBER LAW, PRIVACY, AND POLICY | 02:30 – 03:15 pm

Title: AI is not your Attorney

Dr. Jeff Sallee
Co-chair of the Kentucky Bar Association's AI Committee
University of the Cumberlands

ABSTRACT

Generative artificial intelligence tools are increasingly used for research and decision-making, but relying on AI for legal advice can create significant legal and practical risks. This presentation examines those risks through *U.S. v. Heppner*, a federal case involving a defendant's use of generative AI for legal research, and considers the implications for discovery, attorney-client privilege, and attorney work product. The presentation also explains how large language models generate responses and why AI-produced citations, quotations, and legal authorities must be independently verified. Attendees will learn to recognize hallucinations, bias, and common logical fallacies in AI-generated content and will leave with practical guidance for critically evaluating AI responses rather than treating them as authoritative legal advice.

BRIEF BIOGRAPHY

Dr. Jeff Sallee is a Senior Cybersecurity Governance Specialist with a Fortune 500 financial services company based in Cincinnati, Ohio. He previously spent a distinguished career with Procter & Gamble before retiring from the organization in 2015.

A U.S. Army veteran, Jeff served in the US Army's Military Intelligence Branch as a German linguist during the Cold War. His responsibilities included monitoring and analyzing voice communications across the Warsaw Pact, providing critical intelligence support during a pivotal period in global security.

In addition to his industry role, Dr. Sallee serves as a full-time Instructor at the University of the Cumberlands, where he teaches across multiple information technology disciplines. He has developed several university courses, including *Ethics and Artificial Intelligence*, and serves on the university's Cybersecurity and Information Systems Advisory Board.

A recognized authority in cybersecurity, artificial intelligence, and technology law, Dr. Sallee frequently presents and publishes on these topics for the Kentucky Bar Association and regional bar associations throughout Kentucky, Ohio, and Virginia. His professional cybersecurity certifications include the Certified Information Systems Security Professional (CISSP), the Certified Information Systems Auditor (CISA), and the Certified Cloud Security Professional (CCSP).

Throughout his legal career, Dr. Sallee has held numerous leadership positions within the Kentucky Bar Association, including serving as Chair of the Small Firm and Solo Practitioner Section, the Committee on the Practice of Law, and the Artificial Intelligence Committee. He has also contributed his expertise to multiple governmental and professional committees focused on legal, technology, and cybersecurity issues.

Session: DIGITAL FORENSICS | 03:30 – 04:15 pm

Title: AI's Impact on Cybersecurity and (Both Digital and Conventional) Forensics

Dr. Jeremy Straub
Associate Professor of Cybersecurity and AI
University of West Florida

ABSTRACT

This presentation will cover advancements in AI and how this is changing investigation. Topics like evidence integrity, falsified evidence impact and detection and the use of AI in investigations will be covered.

BRIEF BIOGRAPHY

Dr. Jeremy Straub is an associate professor at the University of West Florida Center for Cybersecurity and AI. He holds two B.S. Degrees (business and IT), two M.S. Degrees (MBA and MS in Computer Systems and Software Design), a Ph.D. in Scientific Computing and Graduate Certificates in Business and Space Studies. He has, previously, been awarded over $7 million in grant and contract funding, much of this as PI. His work has been supported by agencies including the U.S. Department of Defense, National Aeronautics and Space Administration, National Institute of Health, National Security Agency and the National Science Foundation. His work has appeared in national news outlets including CNN, Forbes, USA Today, CBS, Newsweek, Fox News and The Hill.

Session: AWARDS AND CLOSING | 15 mins

Dr. Abdul R. Yarali

President and Founder of KY-CAE

BRIEF BIOGRAPHY

Professor Abdul R. Yarali is the President and Founder of the Kentucky Center of Academic Excellence (KY-CAE), a collaborative initiative uniting colleges and universities across Kentucky that offer cybersecurity, forensics, and computing programs. Through this forum, institutions work together on joint cybersecurity initiatives, advancing education, research, and statewide awareness in this critical field.

As the founder of the Kentucky Cybersecurity and Forensics Conference (KCFC), Dr. Yarali has played a pivotal role in raising cybersecurity awareness throughout the Commonwealth. His leadership has established KCFC as a premier annual event that brings together academia, industry, and government to address emerging cybersecurity challenges.

Dr. Yarali is an internationally recognized researcher and educator in engineering, wireless communications, cybersecurity, and networking. He earned his B.S., M.S., and Ph.D. in Electrical Engineering from the University of Florida, George Washington University, and Virginia Polytechnic Institute and State University (Virginia Tech), respectively. His professional career began in the wireless mobile communications industry, where he served as a technical advisor and engineering director.

He has served as a keynote speaker, panelist, chair, and organizer for national and international conferences and workshops on topics such as wireless communications, cybersecurity, drones, and small satellites. His research collaborations include projects with the National Institute of Hometown Security (NIHS), funded by the Department of Homeland Security, and the NASA Kentucky Drone project.

Throughout his career, Dr. Yarali has combined vision, leadership, and technical expertise to achieve significant success for universities, companies, organizations, and students alike. His work continues to inspire innovation and collaboration in the fields of engineering, wireless communication, and cybersecurity.

2026 Kentucky Cybersecurity and Forensics Conference

STUDENT POSTER ABSTRACTS

Friday, October 30, 2026

Harold R. Benson Research and Demonstration Farm

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Student Poster Session

Time: 12:00 pm – 1:00 pm

JUDGES

Elwyn Rainer II

James Meece

Sheri Donahu

UAV-Based Multispectral Remote Sensing and Machine Learning for Soybean Yield Prediction

Susma Adhikari
Graduate Student and Research Assistant
Kentucky State University
Session: Student Poster Session | **12:00 pm – 1:00 pm**

ABSTRACT

Accurate in-season soybean yield prediction can improve precision agriculture and nitrogen management. Because non-nodulating soybean lacks biological nitrogen fixation, its canopy response and yield formation depend more directly on soil and fertilizer nitrogen. This field study evaluated UAV-derived multispectral vegetation indices (VIs) and machine-learning models for predicting grain yield in two non-nodulating cultivars. Imagery was acquired on six post-planting dates during 2025 using a DJI Matrice 300 RTK with a MicaSense five-band sensor. Eleven VIs were used in six models: ordinary least squares, Ridge, Lasso, random forest, gradient boosting, and support vector regression. Performance was assessed by five-fold cross-validation using R^2, root mean square error and mean absolute error. The full seed stage (R6) produced the strongest yield signal. Chlorophyll Index Green showed the strongest single-index relationship with yield ($R^2 = 0.67$), followed by GNDVI, whereas NDVI was less predictive. With all indices, Ridge regression and random forest achieved the highest cross-validated performance (CV $R^2 \approx 0.50$), while ordinary least squares performed worst. Feature importance identified CIG, NDVI, GNDVI, and LCI as the leading predictors at R6. These results show that well-timed UAV multispectral sensing combined with regularized or ensemble models can support in-season yield assessment in nitrogen-dependent soybean systems.

Hardware Acceleration of Post-Quantum Cryptography: Optimizing ML-KEM on RISC-V Architectures

Travis Blauser
System Administrator
Kentucky State University
Session: Student Poster Session | **12:00 pm – 1:00 pm**

ABSTRACT

The impending arrival of cryptographically relevant quantum computers threatens legacy public-key infrastructure, prompting NIST's standardization of the Module-Lattice-Based Key-Encapsulation Mechanism (ML-KEM). However, executing ML-KEM's modular polynomial arithmetic and Keccak hashing on general-purpose CPUs causes severe pipeline stalls and excessive energy consumption. This study investigates how custom hardware modifications within the open-source RISC-V architecture can accelerate ML-KEM execution while maintaining power efficiency and physical side-channel security. Using a comparative quantitative methodology, this paper evaluates four architectural paradigms (software baseline, decoupled coprocessor, domain vector processor, and tightly coupled Instruction Set Extensions) across execution latency, silicon area overhead, dynamic power, and side-channel resilience. Empirical benchmark analysis reveals that tightly coupled ISEs achieve a 65.9x speedup over software baselines while requiring only 3,800 additional lookup tables and drawing $4.07\ \mu J$(4.07 u J) per operation. This represents an 84% energy reduction compared to vector units and a 90% reduction compared to decoupled coprocessors. These findings demonstrate that tightly coupled instruction-level acceleration offers the most efficient, secure framework for deploying post-quantum cryptography in resource-constrained embedded systems.

UAV-based time-series images to assess vegetation health across multiple planting dates in soybean

Anjan Timilsina, Anuj Chiluwal and Buddhi Gyawali
Kentucky State University
Session: Student Poster Session | **12:00 pm – 1:00 pm**

ABSTRACT

Soybean (Glycine max (L.) Merril) is one of the most popular agronomic crops for its high demand in the food and feed industry throughout the world. Improvement in soybean seed yield and seed composition is critical to producing high-quality feed. However, several factors such as environmental, genetic, and agronomic practices have pivotal roles in determining the quality and quantity of seed yield. The ever-changing climatic patterns might have shifted the optimal planting dates for soybean in Kentucky and across the USA. Therefore, it is essential to study several planting dates covering the early period to late season soybean planting time. Regular crop monitoring might provide information about critical growth and development periods across different planting dates. However, a manual phenotyping-based approach is time-consuming, labor-intensive, and often error-prone in a large area. Thus, unmanned aerial vehicles (UAVs) using drones and advanced sensors are popular for High-throughput data collection, which offers an efficient and non-destructive method for monitoring crop growth and estimating morphophysiological traits. Thus, a study is conducted to assess vegetation health using multiple vegetation indices across multiple planting dates and cultivars in soybean. The study will assess the relationship between manually measured yield-related parameters like plant height, leaf area index, stomatal conductance, and leaf chlorophyll content and vegetation indices derived from drone-based images. Ultimately, the study will identify the most effective vegetation indices for predicting key morphological and physiological parameters as well as yield in soybean across multiple planting dates. Furthermore, the study will also identify the optimal planting date of soybean in Kentucky based on the relationship between vegetation indices and plant parameters across different planting dates.

Materials Property Optimization for Optical Metasurfaces

Malik J. Thompson[1*], Mulaine Shih[2,3], Eric S. Harper[3]
[1] Kentucky State University, [2] Core4ce, [3] Air Force Research Laboratory
Session: Student Poster Session | **12:00 pm – 1:00 pm**

ABSTRACT

This project is focused on engineering an optical metasurface to create a polarizing beam splitter—an optical device used to manipulate light into two separate beams based on the light's polarization where one is reflected off the metasurface (s-polarized) and the other beam transmitted through the metasurface (p-polarized). Metasurfaces, structures engineered to manipulate light in novel ways simplify this process making it easier to manipulate light on a smaller scale {4}. For this study we're focused on optimizing the coefficients of the Sellmeier Equation for silicon, thereby optimizing its optical properties, and ultimately the performance of the polarizing beam splitter.

Using multi-objective optimization, we want to maintain the highest level of reflectance for the s-polarized light and transmittance for the p-polarized light {3}. Through computational analysis, we're able to optimize the Sellmeier equation through its coefficients as they're defining variables that help us shape the desired refractive index to satisfy our objectives using three designs for the polarizing beam splitter. In addition to describing the effectiveness of the material used to show the optimal performance of reflectance and transmittance of light through the metasurface, optimizing the Sellmeier coefficients also allows us to compare our analysis to real-world models of engineered metasurfaces {1}. The performance is evaluated as an objective function to determine its optimality {2}

Due to the nature of this project, our objectives are competing with one another thus, if there is a less amount of reflectance, there will be a greater amount of transmittance and vice versa. As the optimization continues, a set of optimal solutions that represents the best solutions for both the highest reflected s-polarized light and the highest transmitted p-polarized light becomes present within the data. This set of optimal solutions, known as the Pareto front, are viewed as the tradeoff curve for all possible solutions encountered throughout the optimization process. Complete analysis of the pareto front shows different optimal Sellmeier coefficient combinations for different metasurface designs. Overall, this experimental data can be used to draw conclusions on how to dope silicon to improve performance by comparing it to refractive index curves .

We optimized the Sellmeier coefficients to produce a refractive index that will give us the highest level of transmittance of p-polarized light and reflectance of s-polarized light simultaneously as a polarized beam of light is shone on a metasurface. While our approach is a first-order approximation of doping silicon, this process for optimizing material properties can be expanded to cover other materials for similar applications.

Keywords: Optimization, Polarizing, Sellmeier, Optical, Metamaterials

Model Choice and Cyberattack Detection in an AI-Assisted Biogas Purification Digital Twin

Bradley Coffman, Liang Yu, and Milon Chowdhury
Biological and Agricultural Engineering
Kentucky State University
Session: Student Poster Session | **12:00 pm – 1:00 pm**

ABSTRACT

Digital twins can support monitoring and supervisory control of biogas purification, but manipulated sensor signals may alter their predictions and downstream control decisions. This proposed simulation study examines whether operational model structure influences the detection of sensor manipulation and its consequences for process control in a CO_2 water-wash scrubber.

Three digital-twin formulations will be compared: mechanistic mass-transfer models, statistical models, and hybrid models combining physical principles with neural networks. Constant bias, gradual drift, and fabricated readings will be introduced into the inlet CO_2 sensor signal while the simulated physical inlet conditions remain unchanged. A residual-based integrity monitor will compare model predictions against an assumed-trusted outlet CO_2 measurement, using model-specific detection thresholds calibrated to comparable false-positive rates.

A common AI supervisor will recommend bounded water-flow setpoints during normal operation, with proportional–integral (PI) fallback control following an integrity alert. The simulated plant, attack conditions, supervisory policy, and fallback controller will remain fixed across model comparisons. Detection delay, false-positive rate, outlet CO_2 deviation, and water-flow deviation will be evaluated.

The proposed study aims to determine whether differences in digital-twin model structure influence both the detectability of sensor manipulation and its consequences for automated process control.

Smart Water Cybersecurity: Detecting and Localizing Sensor Anomalies with Spatial and Operational Context

Saiful Islam, Liang You
Biological and Agricultural Engineering
Kentucky State University
Session: Student Poster Session | **12:00 pm – 1:00 pm**

ABSTRACT

Smart water networks require monitoring that identifies suspicious sensor readings while limiting false alarms. This study evaluates whether spatial and operational context improves pressure anomaly detection and sensor localization. Twelve detector variants were compared on the Net3 and KY4 hydraulic models, with three demand realizations per network. Models were trained and calibrated exclusively on normal operation and evaluated using 6,048 controlled sensor-corruption scenarios involving offsets, ramps, frozen readings, and replay. Chronological data separation and paired clean controls supported reproducible comparisons. Adding tank levels, pump states, and pump flows to pressure-based reconstruction increased correct first-alarm sensor localization, counting missed events as failures, from 72.8% to 80.8% in Net3 and from 93.5% to 95.4% in KY4. Meanwhile, normal-hour alarm fractions decreased from 1.29% to 0.93% and from 0.98% to 0.87%, respectively. Spatial-neighbor performance varied by network, and principal component analysis remained competitive, highlighting the importance of strong baselines. These simulation results assume trustworthy operating signals and do not establish malicious intent. The findings support combining process measurements with network information to improve sensor investigation and guide further validation of cybersecurity monitoring for critical water infrastructure.

Effect of Beeswax and Paraffin Wax as Edible Coatings on the Quality and Shelf-Life of Pawpaw Fruit (Asimina triloba)

Pragyan Ghimire*, Jeremy D. Lowe, Sheri B. Crabtree, and Kirk W. Pomper
Land Grant Program, Kentucky State University, College of Agriculture, Health, and Natural Resources, Frankfort, KY 40601
Session: Student Poster Session | **12:00 pm – 1:00 pm**

ABSTRACT

Pawpaw (Asimina triloba) is a native North American fruit with a distinct flavor resembling the taste of banana, pineapple, and mango. The fruit has both fresh consumption and value-added processing potential; however, its high perishability and short shelf-life limit its potential as a fresh market fruit. Pawpaw can be stored only for about 2 to 5 days in non-refrigerated conditions, whereas long-term cold storage results in poor ripening and internal black discoloration. The objective of this study was to evaluate the effectiveness of beeswax, paraffin wax, and a 1:1 (w/w) combination of beeswax and paraffin wax as edible coatings for extending the shelf life of pawpaw fruit, with hot water treatment and an untreated control included for comparison. Fruit was dipped completely in each treatment for approximately 10-15 seconds. Four replicate fruits from the control and hot water treatments were evaluated at day 0 as a baseline, and thereafter for all treatments after 10, 20, 30, 40, and 50 days of cold storage at 5 °C. Data on firmness, sugar content, and weight loss were recorded throughout the storage period. Fruit firmness decreased in all treatments over the storage period. However, fruits treated with a combination of beeswax and paraffin wax showed higher firmness on day 10. After that, no significant differences were found. The weight loss percentage was significantly lower in coated fruit throughout the storage period. There was no difference in fruit Brix (%) levels across all the treatments. Edible coatings, particularly the combination of beeswax and paraffin wax, maintained fruit quality during cold storage by reducing weight loss, but did not preserve firmness.

TransFAE: Transformer-Enhanced Fractal Autoencoder for Representative and Diverse Feature Selection and Classification

Sulav Poudel
Kentucky State University
Session: Student Poster Session | **12:00 pm – 1:00 pm**

ABSTRACT

High-dimensional datasets often contain irrelevant, redundant, and strongly correlated variables that increase computational cost, reduce interpretability, and may limit the generalization of downstream learning models. Existing unsupervised feature-selection methods address this problem through manifold preservation, sparsity, reconstruction, differentiable subset selection, and redundancy control. Fractal Autoencoders (FAE) further improve unsupervised feature selection by combining a globally weighted reconstruction network with a local top-k reconstruction subnetwork, encouraging the selected variables to remain representative and diverse. However, FAE-style reconstruction does not explicitly model contextual and higher-order inter-feature relationships through self-attention. We propose TransFAE-U, a Transformer-enhanced fractal autoencoder for unsupervised feature selection that integrates FAE-style global-local reconstruction with feature-axis self-attention. The proposed model learns a single dataset-level importance score for each original variable and constructs two shared reconstruction pathways: a softly weighted global branch and an exact top-k local branch. Crucially, feature selection is performed on the original variables before self-attention, and the values of unselected features are replaced with learned mask representations, preventing omitted variables from influencing selected representations through prior contextual mixing. A shared Transformer subsequently models nonlinear and contextual inter-feature dependencies, while a cross-feature reconstruction objective requires the selected subset to recover both the complete input and, specifically, the omitted variables. An explicit redundancy regularizer further discourages highly correlated selections. TransFAE-U therefore targets four desirable properties of feature selection: representativeness, compactness, diversity, and interaction-aware information preservation. The framework will be assessed across heterogeneous tabular, biomedical, genomic, image, sensor, speech, text, and hyperspectral datasets using reconstruction error, classification accuracy, AUROC, F1-score, feature redundancy, selection stability, computational efficiency, and multiple feature-subset sizes. We hypothesize that pre-attention original-feature selection combined with Transformer-based interaction modeling will produce compact, stable, and informative feature subsets that preserve the structure and predictive utility of high-dimensional data more effectively than existing unsupervised feature-selection approaches.

From Weather Warnings to Community Resilience: The Role of Extension and the Future of AI-Enabled Preparedness

Shristi Adhikari
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College of Agriculture, Health, and Natural Resources
Kentucky State University
Session: Student Poster Session | **12:00 pm – 1:00 pm**

ABSTRACT

Extreme weather events pose increasing challenges to agricultural communities, highlighting the importance of effective preparedness, risk communication, and community resilience. This study explores the role of Extension professionals in supporting farmers' preparedness for extreme weather events in Kentucky. Guided by Social Learning Theory, the study examines how Extension facilitates preparedness-related learning through observation, knowledge sharing, and social interaction. A qualitative research approach was employed, using semi-structured interviews with 14 Extension professionals affiliated with Kentucky State University and the University of Kentucky. Interview data were analyzed thematically, resulting in four major themes: (1) multifaceted role of Extension in community resilience, (2) increase in severity and frequency of extreme weather events, (3) preparedness efforts across Extension programs, and (4) challenges and capacity needs for effective community outreach. Together, these themes provided insights into Extension professionals' perspectives on weather-related challenges, their roles in supporting farmers and communities, preparedness efforts across Extension programs, and the constraints affecting effective outreach. Looking ahead, artificial intelligence (AI) presents a potential opportunity to complement Extension-led efforts through localized weather-risk communication, scenario-based preparedness planning, and accessible, multilingual guidance

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BRIEF BIOGRAPHY

Bob Stamper is the Owner and CEO of iSAFE Complete, where he has spent more than 25 years helping protect businesses across Kentucky. He holds the Certified Information Systems Security Professional (CISSP) certification, serves as an ambassador for the Richmond Chamber of Commerce, Commerce Lexington, was named the 2023 Ambassador of the Year in Richmond and was recognized as one of the Top 25 Leaders in Madison County in 2025. Internationally recognized for excellence in leadership, service, and growth, iSAFE Complete was named a finalist in the MSP Titans of the Industry awards in 2025.